

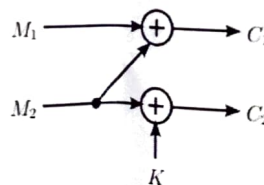
Roll number and Name: _____

Note:

1. The test has 2 parts. Both the parts are for 20 marks each.
2. Part A is meant to have short answers. I do not expect you to write more than one small paragraph in any question. Each question in this part carries 2 marks.
3. Part B has 5 questions of 4 marks each.
4. Write briefly and precisely. Do not waste your and my time with long-winded answers.
5. All answers should have valid reasoning. Writing the final answer, even when correct, does not lead to any marks.

Part A

1. Prove that (a) the shift cipher, and (b) the substitution cipher can be easily broken using a known-plaintext attack. How much known plaintext is needed to completely recover the key for each of these ciphers? (Assume that the underlying language is English with only lowercase letters being used.)
2. Fill in the blanks:
 - (a) _____ is a way to create a stream cipher out of a block cipher.
 - (b) A symmetric-key algorithm for ensuring that a message has not been tampered with is called _____.
 - (c) _____ is a widely used, standardized and secure hash function.
 - (d) A value used in symmetric key cryptography to ensure that a new session that transmits the same text as a previous session does not result in identical ciphertext is called _____.
3. Let the encryption of a $2n$ -bit message $M = (M_1, M_2)$ be defined by the following figure.



Is this encryption function perfectly secure? Why or why not?

4. If the second block of a long ciphertext gets corrupted and exactly two bits in this block are sent erroneously, what effect will it have on the decryption side if the CBC mode of operation is being used with AES?

5. State (no need to prove) the key complementation property of DES. What is the implication of this property on the security of DES?
6. An RSA cryptosystem has public key $n = 35$ and $e = 7$. Find the decryption exponent.
7. A bank manager uses Shamir's (2,3) secret sharing scheme, working modulo a prime 101. The shares created by the manager are (1,13), (3,12) and (2,*) where * represents unreadable data. What should be the value of * so that the scheme works correctly.
8. Given $n = 323737$ and $\phi(n) = 322596$, factorize n .
9. Let $n = p_1 \times p_2 \times p_3$ where p_i are prime numbers. How many square roots of 1 exist modulo n ? Why?
10. State the encryption and decryption functions of the Goldwasser-Micali cryptosystem for the modulus $n = 35 = 5 \times 7$. Show the encryption of a bit $b=0$ for this modulus, and show how to perform the decryption of the generated ciphertext.

Part B

1. A b -bit block cipher with key K , denoted by $E_K(\cdot)$, is used to construct a hash function $h(\cdot)$ in the following way:

First make the length of the input message m a multiple of b , by appending a single 1 bit and then enough 0's (i.e. the 10* padding). Let the resultant message have n blocks and be denoted by $m_0 || m_1 || \dots || m_{(n-1)}$ where each m_i is b bits.

Run the following algorithm:

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 $c = E_{m_0}(m_0)$ 
for( $i = 1$  to  $n - 1$ ) {
     $d = E_{m_0}(m_i)$ 
     $c = c \oplus d$ 
}
Return  $h(m) = c$ 

```

Show that this is not a secure cryptographic hash function (by exhibiting an attack).

2. The ElGamal signature scheme is defined by the following 3 algorithms:

Algorithm 1 Key Generation

- 1: Choose a large prime p randomly.
- 2: Choose a generator α of Z_p^* randomly.
- 3: Choose a random integer $d \in \{2, 3, \dots, (p - 2)\}$.
- 4: Compute $\beta = \alpha^d \mod p$.
- 5: **Public key** = (p, α, β) . **Secret key** = d .

Algorithm 2 Signature generation on message m

- 1: Choose a random number $k \in \{0, 1, \dots, (p - 2)\}$ such that $\gcd(k, p - 1) = 1$.
- 2: Compute parameters (r, s) as follows:
- 3: $r = \alpha^k \mod p$
- 4: $s = (m - d \cdot r)k^{-1} \mod (p - 1)$.
- 5: **Return** $\text{Sig}(m) = (r, s)$.

Algorithm 3 Signature Verification

- 1: **Input** message = m , signature = (r, s) . (Public key from Algorithm 1 is also known).
- 2: Compute $t = \beta^r \cdot r^s \mod p$.
- 3: The verification follows from:

$$t \begin{cases} = \alpha^m \mod p & \Rightarrow \text{Valid signature} \\ \neq \alpha^m \mod p & \Rightarrow \text{Invalid signature} \end{cases}$$

- (a) Prove that the Signature scheme works correctly. That is, if a signature is produced by following Algorithms 1 and 2, then it will be verified by Algorithm 3.
- (b) An "existential forgery" against a signature scheme means the ability to create a valid signature on a random message, without knowing the secret key. Prove that one can create an Existential forgery against the ElGamal signature scheme.
3. Let $h : \{0, 1\}^* \rightarrow \{0, 1\}^n$ be a hash function which is preimage resistant, second preimage resistant, and collision resistant. Let $h' : \{0, 1\}^* \rightarrow \{0, 1\}^n$ be the hash function defined as follows:

$$h'(x) = \begin{cases} 0 \parallel x & \text{if } x \in \{0, 1\}^n, \\ 1 \parallel h(x) & \text{otherwise.} \end{cases}$$

What can you say about the following properties of h' ?

- (a) Preimage resistance
 (a) Second preimage resistance
 (a) Collision resistance

Note: $x \parallel y$ denotes concatenation of strings x and y .

4. Let $E : y^2 = x^3 + Ax + B$ be an elliptic curve over a field K .
- (a) Show that any point on E with y -coordinate equal to zero has order 2.
 (b) Suppose $A = 0$. Show that any point on E with x -coordinate equal to zero has order 3.

Note: An element $a \in$ a group G with additive identity e is said to have order m if

$$a + a + \dots (m \text{ times}) = e.$$

5. Consider the following protocol:

Input: RSA modulus n and a value $x \in QNR_n$.

The prover P knows the factorization of n and wants to prove that x is in QNR_n . (Recall that QNR_n is the set of elements which are quadratic non-residue modulo n . That is, there does not exist a z such that $z^2 \equiv x \pmod{n}$).

1. The verifier V forms a challenge value as follows: choose $c \in \{0, 1\}$, choose random $r \in Z_n^*$. V sends $y = r^2 x^c$ to P .
2. The prover uses the factorization of n to determine whether $y \in QR_n$. If so, let $c' = 0$, otherwise let $c' = 1$. Send c' to V .
3. V checks that $c' = c$. If so, V accepts, otherwise V rejects.

Prove the following regarding this protocol:

- (a) The protocol is sound. (Recall the definition of "soundness" for a zero-knowledge proof).
- (b) The protocol is not zero-knowledge. That is, a cheating verifier can learn "something" by interacting with an honest prover. (Decide what is "something" by thinking along the definition of a zero-knowledge proof. If the proof is ZK then the entire process of the proof could be simulated by someone not having access to the secret information with the prover. If the verifier can learn anything more than what a simulator can, then this is "something").

*** End of the test paper ***